

WasteWise — Product Requirement Document (PRD) & Platform Blueprint

Version: 2.0.0
Target Platform: WasteWise AI-Driven Circular Waste Intelligence System
Tech Stack: React (TypeScript), Tailwind CSS, Framer Motion, Leaflet + OSM, Express.js (Node.js), MongoDB (Atlas), OSRM Routing Engine

Table of Contents

- 1. [Executive Summary & Vision](#)
- 2. [User Personas & Role-Based Access Control](#)
- 3. [Page-by-Page Product Architecture](#)
- 4. [Deep-Dive Feature Specifications](#)
 - [4.1 OSRM Road Navigation & Live Tracking](#)
 - [4.2 Society & Community Leaderboard](#)
 - [4.3 Real Interactive Rewards & Voucher System](#)
 - [4.4 AI Trash & Waste Classifier](#)
 - [4.5 Garbage Reporting & Complaint Dispatch](#)
 - [4.6 Collector & Admin Dashboards](#)
- 5. [Specifications for Placeholder / Missing Pages & Buttons](#)
- 6. [Step-by-Step User Guide \(How to Test & Use\)](#)
- 7. [API Route Mapping & Security Model](#)

1. Executive Summary & Vision

WasteWise is a full-stack, AI-driven circular waste intelligence platform designed for urban sanitation authorities (such as MCD in Delhi) and urban residents. It bridges the gap between citizens, waste collectors, and municipal administrators through:

- **Real-world street navigation** (powered by OpenStreetMap + OSRM) instead of straight-line coordinates.
- **AI-powered waste segregation** to educate citizens on dry, wet, hazardous, and e-waste disposal.
- **Community gamification** with society/RWA rankings, Eco-Scores, and digital voucher redemption.
- **Real-time operations tracking** for collection vehicles and grievance resolution.

Complete Technology Stack & Architecture

1. Frontend Technologies

- **Core Framework:** React 18 with TypeScript
- **Build Tool & Bundler:** Vite 5
- **Styling & UI Tokens:** Tailwind CSS + Vanilla CSS tokens + Glassmorphic dark/light aesthetics
- **UI Components:** Shadcn UI component primitives
- **Icons & Visuals:** Lucide React (`lucide-react`)
- **Animations & Micro-interactions:** Framer Motion (`framer-motion`)
- **Mapping & GIS Engine:** Leaflet + OpenStreetMap Tiles
(`https://tile.openstreetmap.org/{z}/{x}/{y}.png`)

- **Routing & State Management:** React Router DOM v6 + React Context API (`AuthContext`)

2. Backend & API Services

- **Runtime:** Node.js
- **Web Framework:** Express.js
- **Authentication & Security:** JWT (JSON Web Tokens) with 64-character secret key + bcryptjs password hashing
- **File Upload Handler:** Multer (local disk storage for citizen photo reports)
- **API Proxy Architecture:** Express router proxy for OSRM navigation service (`/api/routing/route`) to prevent external API key leaks in client bundle
- **Environment Configuration:** `dotenv` configuration management

3. Database & Data Modeling

- **Database Engine:** MongoDB Atlas (Cloud NoSQL Cluster)
- **Object Data Modeling (ODM):** Mongoose
- **Geospatial Features:** 2dsphere GeoJSON Point indexing for real-time location tracking & distance calculation
- **Data Models:** `User` , `WasteCollection` , `Complaint` , `RewardTransaction` , `Route` , `Grievance`

4. Routing & Navigation Infrastructure

- **Routing Machine:** OSRM (Open Source Routing Machine) Driving Engine (`/route/v1/driving/...`)
- **Route Optimizer:** Custom 2-layer greedy nearest-neighbor ordering + OSRM road polyline generator
- **Polyline Decoder:** Dynamic GeoJSON coordinate decoder for Leaflet street rendering

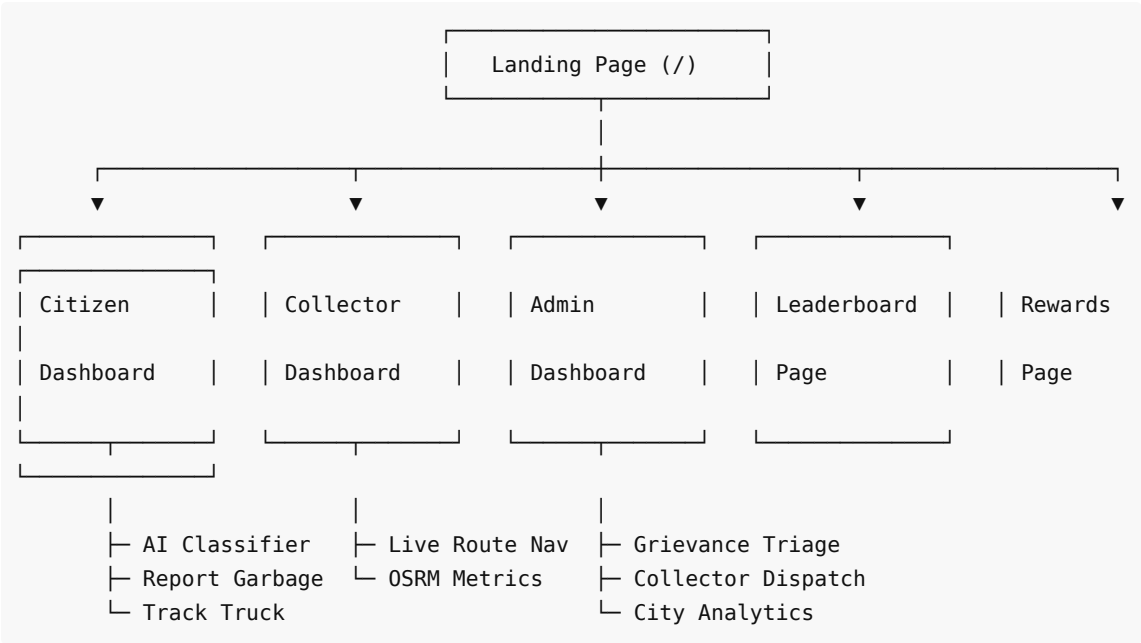
5. AI & Machine Learning Integration

- **Waste Classification Model:** Google Gemini 2.0 / SWC AI model integration for real-time image analysis, dry/wet/e-waste identification, and bin color recommendations

2. User Personas & Role-Based Access Control

Role	Access Level	Primary Objectives	Default Test Login
Citizen	Public / Logged In	Report uncollected garbage, track collection trucks, classify waste with AI, earn & redeem rewards, view society rank.	<code>citizen@wastewise.com / citizen123</code>
Collector	Authenticated	View daily assigned collection route, navigate using turn-by-turn road geometry, mark stops completed, track shift metrics.	<code>collector@wastewise.com / collector123</code>
Admin	Superuser	Monitor city-wide waste statistics, assign collection routes to collectors, triage citizen complaints, manage users.	<code>admin@wastewise.com / admin123</code>

3. Page-by-Page Product Architecture



Detailed Page Catalog

1. Landing Page (`Index.tsx` / `/`)

- **Purpose:** High-impact hero portal showcasing platform value, live statistics, quick links to dashboards, and CTA buttons to login or register.
- **Key Components:**
 - Hero Section with glassmorphic cards and floating animations.
 - Quick Feature Matrix (AI Classifier, Live Route Tracking, Leaderboards, Rewards).
 - Public Impact Counter (Tons of waste collected, active societies, citizen reports resolved).

2. Citizen Dashboard (`CitizenDashboard.tsx` / `/citizen`)

- **Purpose:** Central workspace for residents.
- **Key Features:**
 - Quick action widgets: **Report Garbage**, **Identify Waste with AI**, **Track Nearby Truck**, **Redeem Points**.
 - Personal stats card: Available Eco Points, current Level, activity history.
 - Live complaint status monitor.

3. Collector Dashboard (`CollectorDashboard.tsx` / `/collector`)

- **Purpose:** Command screen for waste vehicle drivers and workers.
- **Key Features:**
 - **Interactive OSRM Live Map:** Real road-following truck movement, pickup pins, and depot locations.
 - **Navigation Card:** Live ETA, distance remaining (km), next pickup location address, and completed stop count.
 - **Route Controls:** "Start Navigation", "Mark Pickup Complete", "Pause Shift".

4. Admin Dashboard (`AdminDashboard.tsx` / `/admin`)

- **Purpose:** Command center for municipal administrators (e.g., MCD officials).
- **Key Features:**
 - City Sanitation Overview (Total active complaints, open vs. resolved cases, active trucks).
 - Grievance Triage List: View citizen photo reports, GPS coordinates, priority level, and assign to nearby collectors.
 - Collector Management Table: Vehicle numbers, assigned zones, shift status.

5. Leaderboard Page (`LeaderboardPage.tsx` / `/leaderboard`)

- **Purpose:** Multi-level competitive community ranking portal.
- **Key Features:**
 - **Top 3 Animated Podium** (Gold, Silver, Bronze badges).
 - **3 Tab Views:** Citizens, Societies/RWAs, Collectors.
 - **Society Eco-Score Bar:** Visual efficiency rating for residential societies.
 - **Search & Filter:** Find your RWA or username instantly.

6. Rewards Center (`RewardsPage.tsx` / `/rewards`)

- **Purpose:** Digital wallet for earning & redeeming points.
- **Key Features:**
 - **Daily Check-in Bonus (+25 Pts):** Single-click daily reward claim.
 - **Digital Voucher Catalog:** 6 options (DJB Water Bill, BSES Power Bill, Jio/Airtel Recharge, Amazon Gift Voucher, DMRC Metro Pass, Flipkart Green Card).
 - **Voucher Generator Modal:** Generates instant codes (e.g. `WW-WATER-8X92K`) with a one-click copy button.
 - **Points History Filters:** All / Earned (+) / Redeemed (-).

7. AI Waste Classifier (`ClassifierPage.tsx` / `/classifier`)

- **Purpose:** AI model for instant garbage identification & bin recommendations.
- **Key Features:**
 - Photo upload & Live camera capture interface.
 - AI Output Card: Waste type (Plastic, Organic, E-Waste, Glass, Paper), Recyclability %, Bin color recommendation (Green, Blue, Yellow, Black).

8. Community Page (`CommunityPage.tsx` / `/community`)

- **Purpose:** Neighborhood feed & local cleanup drives.
- **Key Features:**
 - Community posts feed, upcoming waste drives, event RSVP buttons, and RWA announcements.

4. Deep-Dive Feature Specifications

4.1 OSRM Road Navigation & Live Tracking

- **Frontend File:** `src/components/LiveMap.tsx` , `src/services/osrmRouting.ts` , `src/services/routeOptimizer.ts`
- **Backend File:** `backend/routes/routing.js` (`POST /api/routing/route`)
- **How It Works:**
 1. Frontend submits depot, collection coordinates, and processing plant location to `/api/routing/route` .
 2. Backend proxy forwards request securely to the OSRM Driving Engine (`/route/v1/driving/...`).

3. Returns high-precision Polyline geometry (`overview=full&geometries=geojson`).
4. Leaflet renders the dual-polyline: **Dimmed Grey** for completed path, **Emerald Green** for active street route.
5. Smooth interpolation moves the 🚚 truck marker along exact street turns.

4.2 Society & Community Leaderboard

- **Frontend File:** `src/pages/LeaderboardPage.tsx`
- **Backend File:** `backend/routes/leaderboard.js` (`GET /api/leaderboard/societies` , `/citizens` , `/collectors` , `/overview`)
- **How It Works:**
 1. Backend aggregates MongoDB `User` documents grouped by `profile.society` .
 2. Calculates total members, total points, average level, and total reports + collections.
 3. Computes the **Eco-Score** (composite rating of points efficiency and community activity).
 4. Renders top 3 in an animated 3D podium and highlights the logged-in user with a **YOU** badge.

4.3 Real Interactive Rewards & Voucher System

- **Frontend File:** `src/pages/RewardsPage.tsx`
- **Backend File:** `backend/routes/citizen.js` (`POST /api/citizen/daily-claim` , `POST /api/citizen/redeem` , `GET /api/citizen/rewards/:userId`)
- **How It Works:**
 1. Citizen clicks **Daily Check-in** to claim +25 Pts once every 24 hours.
 2. When selecting a voucher (e.g. DJB Water Bill - 500 Pts), the system validates point balance.
 3. Generates a random unique voucher code (e.g. `WW-WATER-7Y82A`) and updates MongoDB points balance.
 4. The code is displayed in a popup with a "Copy Code" button and saved in the user's transaction history.

5. Specifications for Placeholder / Missing Pages & Buttons

The following pages and buttons currently exist as placeholders or require expanded UI views. Here is their full specification for production:

1. User Profile & Settings Page (`/profile` — `ProfilePage.tsx`)

- **Current State:** Basic form showing name and email.
- **Specification for Complete Feature:**
 - **Society / RWA Selection Dropdown:** Allow citizens to select or change their registered RWA society (e.g., *Green Park RWA*, *Dwarka Sec-7 RWA*).
 - **Zone & Address Manager:** Input street address, landmark, and ward number for accurate garbage pickup.
 - **Vehicle Details (For Collectors):** Input vehicle registration number (e.g. `DL-01-AB-1234`), capacity (in tons), and assigned depot.
 - **Notification Preferences:** Toggle SMS / Email / Push notifications for truck arrival alerts and complaint status updates.

2. Live Tracking Standalone Page (`/live-tracking` — `LiveTrackingPage.tsx`)

- **Current State:** Minimal placeholder wrapper.
- **Specification for Complete Feature:**

- **Full-Screen GIS Map:** Leaflet map with all active MCD collection trucks in the zone.
- **ETA Calculator Widget:** Citizens input their address/society and get real-time ETA for the nearest collection vehicle.
- **Truck Details Drawer:** Click on any truck icon to see driver name, vehicle number, current speed, and route progress.

3. Contact & Support Page (/contact — `ContactPage.tsx`)

- **Current State:** Basic contact form.
- **Specification for Complete Feature:**
 - **Helpline Directory:** Emergency contacts for MCD Control Room, Swachh Bharat Helpline (1969), and Zonal Nodal Officers.
 - **Direct Ticket Submission:** Submit platform bugs or feedback directly to the support backend.

4. Privacy Policy & Terms (/privacy — `PrivacyPolicyPage.tsx`)

- **Current State:** Static text layout.
- **Specification for Complete Feature:**
 - Data protection terms regarding GPS location usage, photo submission privacy, and reward point security.

6. Step-by-Step User Guide (How to Test & Use)

Scenario A: Testing as a Citizen

1. Go to `http://localhost:8080/login` .
2. Login with: `citizen@wastewise.com` / `citizen123` .
3. **Daily Reward Claim:** Go to **Rewards Center** (`/rewards`), click **Daily Check-in (+25 Pts)**. Watch your points balance update!
4. **Voucher Redemption:** Click on **DJB Water Bill (500 pts)**, click **Confirm Redeem**, and copy your generated voucher code!
5. **Leaderboard:** Go to **Leaderboard** (`/leaderboard`). Click on the **Societies / RWAs** tab to view *Green Park RWA, Dwarka Sec-7 RWA*, etc.
6. **AI Waste Classifier:** Go to **AI Classifier** (`/classifier`), upload a photo of garbage or plastic, and inspect the AI classification and bin recommendation.

Scenario B: Testing as a Waste Collector

1. Login with: `collector@wastewise.com` / `collector123` .
2. Navigate to **Collector Dashboard** (`/collector`).
3. View the **Interactive OSRM Live Map** with the real road-following route line.
4. Click **Start Route Navigation** to watch the truck move along streets with dynamic distance (km) and ETA updates.

Scenario C: Testing as an MCD Admin

1. Login with: `admin@wastewise.com` / `admin123` .
2. Navigate to **Admin Dashboard** (`/admin`).
3. View city-wide complaint statistics, active collection vehicles, and triage citizen reports.

7. API Route Mapping & Security Model

Method	Endpoint	Purpose	Access
POST	/api/auth/login	User login & JWT issuance	Public
POST	/api/routing/route	OSRM Road navigation proxy	Internal Proxy
GET	/api/leaderboard/societies	Aggregated RWA leaderboard	Authenticated
GET	/api/leaderboard/citizens	Top citizen rankings	Authenticated
GET	/api/leaderboard/collectors	Top collector rankings	Authenticated
GET	/api/citizen/rewards/:userId	Get points & transactions	Authenticated
POST	/api/citizen/daily-claim	Claim daily +25 pts bonus	Authenticated
POST	/api/citizen/redeem	Redeem voucher & generate code	Authenticated
POST	/api/citizen/report	Upload garbage report & photo	Authenticated